Moses Kolleh Sesay

Sustainability & Climate Analyst

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Open to sustainability, climate-risk, ESG and sustainable-AI roles and consulting — EU, remote-friendly

PROFILE

I turn environmental data into decisions leaders can act on.

Geologist by training, sustainability analyst by conviction. I've drilled for water in Sierra Leone, modelled river pollution for the world's coasts, supported the UN's disaster-risk work — and now I'm teaching AI to tread lightly on the planet.

EXPERIENCE

Researcher — Sustainable Artificial Intelligence Sept 2025 – Present
Digital Society School, Amsterdam University of Applied Sciences

- Developing a framework and working prototype that weighs the benefits of generative AI against its environmental costs
- Mapping the Scope 2 and Scope 3 footprint of AI workloads — electricity, hardware, data-centre water — against CSRD/ESRS disclosure logic
- Building a decision-support tool for managers setting AI strategy and for the staff who prompt every day
- Anchored in UN SDG 12 (Responsible Consumption) and SDG 13 (Climate Action)

Cohort 7 Member — Sustainability Consulting Accelerator Mar 2025 – Apr 2025
OnePointFive (opf.degree)

- Intensive 8-week programme on Net Zero strategy and decarbonisation
- Frameworks: SBTi, CDP, GHG Protocol, ISO 14000, IFRS S1&S2, TCFD, TNFD
- Hands-on labs: materiality assessments, Scope 1–3 accounting, transition planning

Climate Change Adaptation Intern Jun 2023 – Oct 2023
United Nations Office for Disaster Risk Reduction · Bonn, Germany

- Audited disaster-loss data for Caribbean countries feeding the Sendai Framework Monitor
- Documented 54 global hazard information systems, expanding regional and hazard-type coverage by 25%
- Wrote the disaster and climate-risk factsheet for Trinidad and Tobago, produced for national policy use

Field Operations Manager May 2019 – Sept 2019
Team & Team International · Freetown, Sierra Leone

- Led completion of 14 solar-powered boreholes with storage tanks across Sierra Leone
- Hit a 70% aquifer-detection rate with electrical-resistivity surveys and pumping tests
- Streamlined the borehole design process in Excel and QGIS
- Trained local staff on sustainable water-system maintenance

Operations Supervisor Sept 2018 – Apr 2019
Sierra Drilling and Water Supply Enterprise · Freetown, Sierra Leone

- Managed a drilling team of 23
- Constructed 50 boreholes and rehabilitated 100 hand-dug wells
- Used Excel and QGIS for project management and optimisation

SELECTED PROJECTS

Sustainable AI Framework 2025 – present · Amsterdam, NL
A working decision-support prototype, not a slide deck. The team's prototypes are public on GitHub as Sustainable AI Prototypes, with my role in its README. CHECKABLE FROM OUTSIDE

GAIA — Green AI Assessment 2025 – 2026
The workbook, the web estimator and the reference code give the same answers: 3,984 cross-engine checks pass.
CHECKABLE FROM OUTSIDE

Coastal Water Pollution Dynamics 2023 – 2024 · Wageningen, NL
10,226 sub-basins modelled. A property of the model domain used, not an estimate; the thesis is on request.
NOT CHECKABLE FROM OUTSIDE

Flood-Resilient Wuppertal 2023 · Wuppertal, DE
A complete flood-risk management framework delivered to the municipality. The deliverable is held by the client; nothing here claims it was implemented. NOT CHECKABLE FROM OUTSIDE

UN Disaster Risk Reduction — hazard data Jun 2023 – Oct 2023 · Bonn, DE
54 global hazard information systems identified and documented. A count recorded in the deliverable, which UNDRR holds.
NOT CHECKABLE FROM OUTSIDE

Soft Path Water Management 2020 — 2021 · Changsha, CN

Successfully defended at Hunan University, May 2021. Degree awarded: MSc Industrial Engineering, 2021; the certificate is on request. NOT CHECKABLE FROM OUTSIDE

Groundwater Potential Mapping 2016 — 2017 · Freetown, SL

70% aquifer strike rate in subsequent drilling. From the field records of the boreholes sited with the survey, not a controlled trial. NOT CHECKABLE FROM OUTSIDE

EDUCATION

MSc Environmental Sciences 2021 — 2024

Wageningen University & Research, Netherlands

Human-Environment Interaction · Water Systems & Global Change · Thesis: socioeconomic drivers of river export of pollutants worldwide · Wageningen African Scholarship Programme scholar · GPA 7.8/10 (Dutch scale)

MSc Industrial Engineering 2019 — 2021

Hunan University, Changsha, China

Full Chinese Government MOFCOM Scholarship · Thesis: soft path water management — Freetown case study · Project management, operations research, systems analysis

BSc Honours Geology 2013 — 2017

University of Sierra Leone, Freetown

First Class Honours — graduated top of class · Dissertation: groundwater potential mapping, geophysical approach · Hydrogeology, geophysics, environmental geology

CERTIFICATES

ESG Specialist Program Sep 2024

Corporate Finance Institute

ESG investing, financial strategy and reporting; risk and opportunity analysis

Data Analytics Training Dec 2023

Masterschool

SQL, Python, Excel, Tableau; real-world business cases

Google Advanced Data Analytics Jan 2024

Google — Coursera

Machine learning and regression models; statistical methods and visualisation

Synergizing DRR & Climate Change Adaptation 2023

UN System Staff College

Disaster risk reduction and adaptation, jointly; Sendai Framework in practice

PUBLIC WORK

- Groundwater Investigation Toolkit (Code, 2026) github.com/moseskolleh/WaterProject
- GAIA Framework 2.2 — Green AI Assessment (Framework, 2026) github.com/moseskolleh/GAIA-Framework-
- climatematch-pipeline — climate extremes for West Africa (Code, 2026) github.com/moseskolleh/climatematch-pipeline
- GloBox A/B test analysis (Analysis, 2023) github.com/moseskolleh/A-B-Testing-at-Globox
- Sustainable AI Prototypes (Code, 2025) github.com/moseskolleh/SustainableAIPrototypes
- EcoPrompt Coach — the maintained tool (promptcoach) (Code, 2026) github.com/moseskolleh/promptcoach
- EcoPrompt Coach — this site's edition, with its evidence ledger (Tool, 2026) on the site: carbon-ai.html
- AI carbon factor set — sources, ranges and review dates (Dataset, 2026) on the site: ai-carbon-data.js
- Field Notes (Essays, 2026) on the site: index.html#notes
- Journey map generator — reproducible cartography (Code, 2026) on the site: scripts/generate-journey-map.js

SKILLS

Python	Pandas · NumPy · Matplotlib — proof: Groundwater toolkit: field data in, reports out, on GitHub
QGIS & ArcGIS	geospatial analysis — proof: Groundwater maps that struck water 7 times out of 10, on request; no public GIS project yet
SQL	PostgreSQL · MySQL — proof: GloBox A/B test: the queries and a statistics notebook, on GitHub
Tableau	visual analytics — proof: GloBox A/B test: its README links a Tableau Public dashboard
Machine Learning	regression · classification — proof: Google Advanced Data Analytics certificate; no public project yet
JavaScript	interactive tools — proof: EcoPrompt Coach: engine, web app and extension, on GitHub
Sustainability	ESG Analysis & Integration · GHG Accounting (Scope 1–3) · Life Cycle Assessment · Climate Risk Assessment · Water Resource Management · Disaster Risk Reduction · Carbon Markets · Circular Economy
Frameworks & standards	SBTi · CDP · GHG Protocol · TCFD · TNFD · ISO 14000 · IFRS S1&S2 · SASB · GRI · CSRD